One-to-One Relationships

A one-to-one association (or, more accurately, a one-to-zero-or-one relationship) is implemented using a foreign key in one row in one table to reference at most a single row in another table. A *one-to-one* relationship might exist between orders and invoices: for each order there's at most one invoice.




```
class Invoice < ActiveRecord::Base
  belongs_to :order
  # ...
end
```

```
class Order < ActiveRecord::Base
  has_one :invoice
  # ...
end
```

As the example shows, we declare this in Rails by adding a `has_one` declaration to the `Order` model and by adding a `belongs_to` declaration to the `Invoice` model.

An important rule is illustrated here: the model for the table that contains the foreign key *always* has the `belongs_to` declaration.

One-to-Many Relationships

A one-to-many association allows you to represent a collection of objects. For example, an order might have any number of associated line items. In the database, all the line item rows for a particular order contain a foreign key column referring to that order, as shown in the figure.